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Span and Linear Independence Graded Assignment
Graded Assignment • 20 min

Due Nov 13, 11:59 PM IST

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. **True** or **False**: Every vector space *V* has a finite set *S* which both spans *V* and is linearly independent in *V*. 1 / 1 point

   - True
   - False (selected)

   Nice work

2. **True** or **False**: If *S* spans a subspace of a vector space *V*, then *S* is linearly independent in *V*. 1 / 1 point

   - True
   - False (selected)

   Nice work

3. Determine if the set of vectors $\left\{\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}, \begin{bmatrix}2 & -1\\ 0 & -1\end{bmatrix}, \begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}, \begin{bmatrix}-2 & 1\\ -1 & -1\end{bmatrix}\right\}$ is linearly independent in $\mathbb{R}^{2\times 2}$. 1 / 1 point

   - The set is linearly independent.
   - The set is linearly dependent. (selected)

   Nice work

4. **True** or **False**: If *S* spans a vector space *V*, but *S* is not linearly independent in *V*, then there exists a vector **v** in *S* such that the set difference $S \setminus \{\mathbf{v}\}$ also spans *V*. 1 / 1 point

   - True (selected)
   - False

   Nice work

5. Determine if the set of vectors $\{x,\ x^3 - x,\ x^5 - x^3,\ x^7 - x^5,\ \dots\}$ is linearly independent in **P**. 1 / 1 point

   - The set is linearly independent. (selected)
   - The set is linearly dependent.

   Nice work

6. Determine if the set of vectors $\{x^2 - 1, 2x^2 - 2x + 5\}$ is linearly independent in $\mathbf{P}^2$. 1 / 1 point

   - The set is linearly independent. (selected)
   - The set is linearly dependent.

   Nice work

7. Determine if $f(x) = \cos^2 x$ is in the subspace of $C[0, 2\pi]$ given by $\text{span}\{1, \cos x, \cos(2x)\}$. 1 / 1 point

   - *f* is in the subspace. (selected)
   - *f* is not in the subspace.

   Nice work

8. Determine if $\mathbf{v} = \begin{bmatrix}-2 & -1 & 1\\ -1 & -2 & 0\end{bmatrix}$ is in the subspace of $\mathbb{R}^{2\times 3}$ given by $\text{span}\left\{\begin{bmatrix}3 & 0 & 3\\ 1 & 0 & 3\end{bmatrix}, \begin{bmatrix}-2 & 3 & 0\\ -2 & 1 & 3\end{bmatrix}, \begin{bmatrix}1 & 0 & -1\\ 2 & -2 & 2\end{bmatrix}\right\}$. 1 / 1 point

   - **v** is in the subspace.
   - **v** is not in the subspace. (selected)

   Nice work

9. Determine if the set of vectors $\{1,\ x - 1,\ x^2 - x,\ x^3 - x^2,\ x^4 - x^3,\ \dots\}$ spans **P**. 1 / 1 point

   - The set spans **P**. (selected)
   - The set does not span **P**.

   Nice work

10. **True** or **False**: If *S* is a linearly independent subset of a vector space *V*, but *S* does not span *V*, then there exists a vector **v** in *V* such that the set $S \cup \{\mathbf{v}\}$ is also linearly independent. 1 / 1 point

    - True (selected)
    - False

    Nice work